

Original article

The influence of brewing method on bioactive compounds residues in spent coffee grounds of different roasting degree and geographical originRita Głowacka,^{1*}  Agata Górka,¹ Magdalena Wirkowska-Wojdyła,¹ Rafał Wołosiaś,² Ewa Majewska² & Dorota Derewiaka²¹ Department of Chemistry, Division of Organic and Food Chemistry, Faculty of Food Sciences, Warsaw University of Life Sciences – SGGW, Nowoursynowska 159c, Warsaw 02-776, Poland² Department of Biotechnology, Microbiology and Food Evaluation, Division of Food Quality Evaluation, Faculty of Food Sciences, Warsaw University of Life Sciences – SGGW, Nowoursynowska 159c, Warsaw 02-776, Poland

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Summary The content of bioactive compounds in spent coffee grounds (SGC) was studied. SGC were obtained from *Coffea arabica* beans of different roasting degrees (light and dark) and different geographical origins (Nicaragua, Columbia and Mexico) processed using four brewing methods (mocha, filtered, drip and infusion). The highest caffeine and chlorogenic acid contents were determined in filtered spent coffee extracts. All extracts of light roasted spent coffee grounds showed lower browning index levels in comparison to that from dark roasted spent coffee grounds. Generally, the highest content of total polyphenolic compounds and highest antioxidant capacity were determined in extracts prepared in drip. In conclusion, the results obtained in this study indicate that the spent coffee grounds produced of domestic levels, especially those obtained from filter coffeemaker, could be considered as a good source of natural antioxidants.

Keywords Antioxidant activity, browning index, caffeine, chlorogenic acids, coffee by-products, total phenolics.

Introduction

Coffee is an ancient commodity that is one of the most widely consumed beverages in the world. Coffee beans are generated from the plant *Coffea* L., of which there are more than 70 species. Despite the large varieties of coffee species, only two species are commercially explored worldwide: *Coffea arabica* and *Coffea canephora* (Belitz *et al.*, 2009). Coffee production and consumption levels increase annually, with values reaching 157.4 million bags of 60 kg green bean equivalent (GBE), and 155 million bags of 60 kg GBE consumed in the coffee year 2016/2017 (ICO, 2017).

Chemical composition of green coffee beans depends on the number of parameters such as geographical origin or coffee cherry processing methods used. The roasting process is a crucial step in coffee processing. During this process, organoleptic properties such as flavours, aromas and colour are developed, and affect the quality of the roasted coffee. The roasting process causes changes in the chemical composition and

biological activity of the green coffee beans. In return, roasted coffee beans gain unique and characteristic flavour. Chemical compounds that contribute to sensory properties such as flavour can be later extracted during brewing (at domestic or catering levels) or by industrial solid-liquid extraction processes (at industrial level). In both cases large amounts of coffee waste known as spent coffee grounds (SCG) are produced. It is assumed that from 1 ton of green coffee beans about 650 kg of spent coffee grounds dry matter is generated (Bravo *et al.*, 2012).

Coffee is a rich source of bioactive compounds, which are only partially extracted during brewing process. SCG contain not only residues of beneficial compounds such as phenolic acids and flavonoids, but they also are thought to include some amount of caffeine and tannins which can be toxic to animals or some plant species (Weichselbaum & Buttriss, 2010; Niseteo *et al.*, 2012). Because of that, heedless disposal of SCG can have substantial, negative impact on the environment.

To deal with the ever-growing problem of waste accumulation and environmental degradation caused

*Correspondent: E-mail: rita_glowacka@sggw.pl

by industries, a circular economy (CE) concept was brought to life (Pearce & Turner, 1990). Opposing the traditional linear economy model, CE focuses on renewable energy, recycling of waste materials and extending a product life cycle (Stahel, 2016). Up to year 2015, a total of 114 definitions of CE were created (Kirchherr *et al.*, 2017), each based on at least one of the core CE frameworks – reduce, reuse, recycle, recover (4R framework).

In recent years there have been many concepts of SCG reutilisation. Some of them include: industrial furnace fuel (Silva *et al.*, 1998), soil supplementation for mushroom (e.g. *Flammulina velutipes*) cultivation (Leifa *et al.*, 2000), biofuel production resource (Kondamudi *et al.*, 2008) or adsorbent of metal ions and organic xenobiotics in water (Safarik *et al.*, 2012). Presently, there is a growing need for alternative and natural bioactive compounds to be used instead of synthetic ones. SCG represent potential as a source of natural antioxidant compounds.

So far, to our best knowledge most of the researches focus on the spent coffee grounds obtained directly from the industrial processes such as instant coffee production not the residues generated at the domestic and cafeterias levels. Taking the above into account, the aim of this study was to assess if SCG gained from brewing methods commonly used in households can be a viable source of bioactive compounds such as caffeine and chlorogenic acids.

Materials and methods

Preparation of spent coffee grounds extracts

Light and dark roasted *C. arabica* beans from Columbia, Nicaragua and Mexico were purchased in local Warsaw retail store. Roasted coffee beans were grounded to a medium degree in First Austria Millard FA-5480 (Vienna, Austria) grinder freshly before extraction. For the coffee extraction process, four different methods were used: Moka coffeemaker Ariete Mokina (Florence, Italy), filter coffee machine Lazer (China), dripper and infusion. In all of the brewing methods 6 g·100 cm⁻³ of coffee to water ratio was used. Time of brewing was 4 min.

After the brewing, SCG samples were collected and immediately dried at 103 °C until constant weight was reached. The moisture content of SCG was assessed according to the PN-ISO/11294:2002 methodology. The dried SCG were stored in closed containers in the dark and at room temperature. To obtain antioxidant-rich extracts, 1 g of dried SCG samples was put in 40 cm³ of 60% aqueous methanol. Extraction process was performed at 60 °C in laboratory shaker Elpin Plus type 357 (Lubawa, Poland) with water bath. After

90 min the extracts were centrifuged and filtrated through paper filters.

Analytical methods

Caffeine and chlorogenic acid content

Analysis of caffeine and chlorogenic acids in SCG extracts was performed according to Bravo *et al.* (2012) with modifications. HPLC system consisted of a Shimadzu (Kyoto, Japan) DGU-14A degasser, LC-10AT_{VP} pump, SIL-10AD_{VP} autosampler, SPD-M10A_{VP} diode array detector and SCL-10A_{VP} controller. In the study the analytical column Kinetex C18 (Phenomenex, Torrance, CA, USA) was used. The sample injection volume was 5 µL. The elution was performed in gradient mode using 0.3% acetic acid in water (phase A) and methanol (phase B) as mobile phases, at the flow rate of 0.8 mL min⁻¹. The mobile phase composition was as follows: 0 min 20% B, 15–24 min 50% B, 27–29 min 20% B. Detection of compounds of interest was performed at 272 nm (caffeine) and 326 nm (chlorogenic acids). Caffeine and chlorogenic acid (3-CQA) were identified by comparing received chromatographic data with retention times and UV spectra of external caffeine (Sigma-Aldrich, Poznań, Poland) and 3-CQA (Aldrich, Poznań, Poland) standards. Other chlorogenic acid derivatives (4-CQA, 5-CQA, di-CQA) were identified based on the literature data (Farah *et al.*, 2005; Rodrigues & Bragagnolo, 2013). Typical HPLC chromatograms are shown in Fig. 1.

Browning index

Browning index of SCG extracts was carried out as described by Bravo *et al.* (2012). Briefly 50 µL of the SCG extract sample was mixed with 2 mL of demineralised water. After 2 min the browning index was tested by measuring the absorbance of samples at 420 nm using a Shimadzu UV-1280 spectrophotometer (Kyoto, Japan) against the blank control. Blank samples were prepared in the same way without the addition of the sample.

Antioxidant activity

The ABTS assay was based on the method of Re *et al.* (1999) with slight modifications. Briefly ABTS working solution containing 7.5 mM of ABTS and 2.4 mM of potassium persulfate was allowed to react in the dark at room temperature for 12 h. The solution was later diluted with PBS buffer (pH 7.4) to an absorbance of 0.70 ± 0.20 at 734 nm. After that, 4 mL of resulting ABTS solution was mixed with 40 µL of sample or Trolox standard dilutions and absorbances were recorded at 6 min after mixing. Measured concentrations of Trolox in PBS buffer were used for calibration

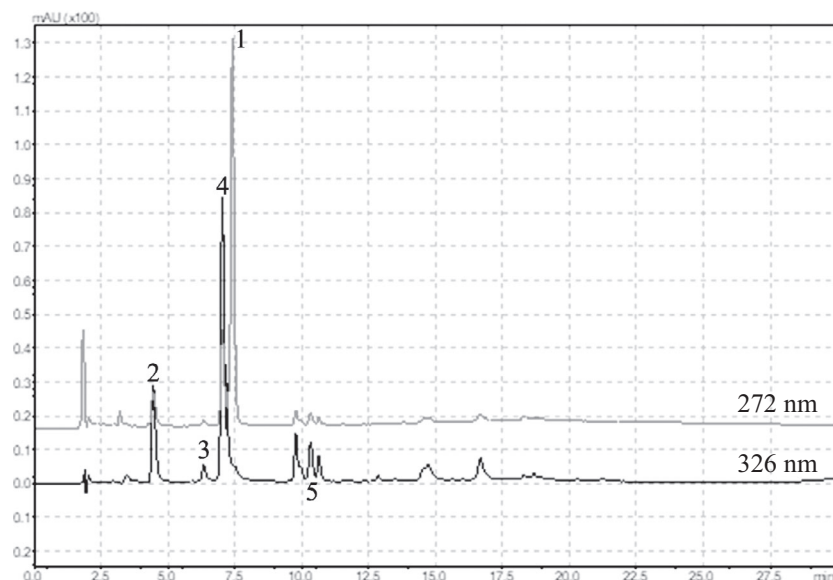


Figure 1 Typical HPLC chromatogram of identified compounds present in spent coffee grounds methanol extracts. (1) caffeine, (2) neochlorogenic acid (5-CQA), (3) cryptochlorogenic acid (4-CQA), (4) chlorogenic acid (3-CQA), (5) dicaffeoylquinic acids (di-CQA).

and the results were expressed as mg of Trolox per gram of SCG dry matter (dm).

Total phenolic content

The total phenolic content of coffee extracts was measured according to Folin-Ciocalteu method (Singleton & Rossi, 1965) with modifications. Briefly, 50 μ L of the sample was diluted to 3.2 mL with distilled water and mixed thoroughly. Then 0.2 mL of Folin-Ciocalteu reagent and 0.6 mL of 20% Na_2CO_3 solution were added to the diluted sample and the solution was mixed again. Solutions were kept in the dark at room temperature for 2 h and total polyphenol content was determined at 765 nm using Shimadzu UV-1280 spectrophotometer (Kyoto, Japan). Gallic acid standard solutions were used for calibration and the results were expressed as mg of gallic acid equivalents per gram of SCG dry matter (dm).

Statistical analysis

Each parameter was analysed in triplicate. Results were shown as the mean $\pm$ standard deviation (SD). Data were analysed using the one-way analysis of variance with Tukey HSD post hoc test ($\alpha = 0.05$). The Statistica 10 (StatSoft Polska, Kraków, Poland) software was used for all calculations.

Results and discussion

Coffee brews can be a rich source of bioactive compounds like caffeine or chlorogenic acids. Extraction of these compounds from roasted coffee beans depends on many different factors like coffee species, roasting degree, grinding degree and brewing methods. But

only a portion of these compounds is transferred to the brew during brewing process.

Caffeine content in SCG extracts reached the values between $2.94 \text{ mg (g dm)}^{-1}$ for spent grounds of light Columbian coffee brewed by drip method and $13.19 \text{ mg (g dm)}^{-1}$ in the case of filtered Nicaraguan dark roasted coffee (Table 1). The obtained results show the influence of brewing method on caffeine content of spent coffee grounds methanol extracts. Highest content of caffeine was present in SCG extracts from filter coffee machine.

In most cases, besides light roasted Columbian and light roasted Mexican coffee, lowest caffeine content was determined in *Mocha* coffeemaker extracts. The results are in agreement with the data provided by Gloess *et al.* (2013) who compared various home brewing methods. Authors obtained brews with caffeine concentration ranging from $4.7 \text{ mg} \cdot 10 \text{ cm}^{-3}$ to $21.0 \text{ mg} \cdot 10 \text{ cm}^{-3}$ in filter coffee and espresso, respectively. Brewing methods based on the principle of percolation e.g. filter coffeemaker, are less effective in extracting caffeine from grounded coffee, compared to other methods e.g. mocha coffeemaker. Therefore spent coffee grounds obtained from percolation brews should be characterised by a higher caffeine content which was confirmed by the result obtained.

The use of HPLC allowed to determine the content of chlorogenic acids present in methanol extracts of coffee grounds. The presence of neochlorogenic (5-CQA), cryptochlorogenic acid (4-CQA), chlorogenic acid (3-CQA) and dicaffeoylquinic acids (di-CQA) was detected in all coffee grounds samples. The content of individual chlorogenic acids evaluated in coffee grounds extracts are presented in Table 2.

Table 1 Caffeine content in spent coffee grounds methanol extracts [mg (g dm)⁻¹]

Brewing	Nicaragua		Columbia		Mexico	
	Light	Dark	Light	Dark	Light	Dark
Drip	6.23 ± 0.01 ^c	6.48 ± 0.12 ^c	2.94 ± 0.06 ^a	6.27 ± 0.12 ^b	4.16 ± 0.11 ^a	5.18 ± 0.01 ^b
Mocha	4.92 ± 0.03 ^a	3.06 ± 0.07 ^a	4.03 ± 0.00 ^b	4.24 ± 0.02 ^a	5.04 ± 0.22 ^b	3.10 ± 0.28 ^a
Filtered	6.34 ± 0.12 ^c	13.19 ± 0.02 ^d	6.67 ± 0.20 ^c	6.30 ± 0.10 ^b	5.81 ± 0.12 ^c	9.48 ± 0.37 ^c
Infusion	5.41 ± 0.01 ^b	3.39 ± 0.04 ^b	4.33 ± 0.14 ^b	4.06 ± 0.06 ^a	5.31 ± 0.04 ^b	4.84 ± 0.13 ^b

Data denoted by the same letters are not statistically different ($\alpha = 0.05$) in terms of brewing method.

Table 2 Chlorogenic acids content in spent coffee grounds methanol extracts [mg (g dm)⁻¹]

	Brewing	Nicaragua		Columbia		Mexico	
		Light	Dark	Light	Dark	Light	Dark
5-CQA	Drip	2.39 ± 0.09 ^b	0.78 ± 0.02 ^c	1.08 ± 0.01 ^a	0.64 ± 0.07 ^b	1.68 ± 0.13 ^a	0.74 ± 0.05 ^a
	Mocha	2.01 ± 0.05 ^a	0.40 ± 0.00 ^b	1.99 ± 0.15 ^c	0.54 ± 0.04 ^b	2.06 ± 0.10 ^b	0.54 ± 0.09 ^a
	Filtered	2.63 ± 0.01 ^c	1.60 ± 0.05 ^d	2.61 ± 0.05 ^d	0.56 ± 0.08 ^b	2.54 ± 0.02 ^b	1.44 ± 0.11 ^b
	Infusion	2.01 ± 0.02 ^a	0.32 ± 0.01 ^a	1.44 ± 0.11 ^b	0.29 ± 0.02 ^a	2.15 ± 0.17 ^c	0.61 ± 0.07 ^a
4-CQA	Drip	0.35 ± 0.02 ^b	0.40 ± 0.02 ^b	0.14 ± 0.03 ^a	0.33 ± 0.02 ^b	0.25 ± 0.00 ^a	0.30 ± 0.02 ^b
	Mocha	0.23 ± 0.01 ^a	0.14 ± 0.03 ^a	0.21 ± 0.02 ^{b*}	0.22 ± 0.01 ^{a*}	0.29 ± 0.01 ^{bc}	0.23 ± 0.02 ^a
	Filtered	0.35 ± 0.01 ^b	0.92 ± 0.01 ^c	0.23 ± 0.01 ^b	0.35 ± 0.01 ^b	0.32 ± 0.02 ^c	0.77 ± 0.01 ^d
	Infusion	0.34 ± 0.02 ^b	0.18 ± 0.04 ^a	0.22 ± 0.01 ^{b*}	0.20 ± 0.03 ^{a*}	0.28 ± 0.01 ^{ab}	0.35 ± 0.01 ^c
3-CQA	Drip	6.29 ± 0.13 ^b	2.12 ± 0.04 ^b	3.01 ± 0.09 ^a	1.55 ± 0.10 ^c	4.57 ± 0.08 ^a	1.98 ± 0.17 ^b
	Mocha	5.23 ± 0.17 ^a	1.08 ± 0.07 ^a	5.14 ± 0.07 ^c	1.27 ± 0.04 ^b	5.98 ± 0.07 ^b	1.35 ± 0.03 ^a
	Filtered	7.25 ± 0.08 ^c	4.21 ± 0.14 ^c	7.51 ± 0.21 ^d	1.73 ± 0.11 ^c	7.30 ± 0.15 ^d	4.50 ± 0.14 ^c
	Infusion	5.81 ± 0.31 ^b	0.91 ± 0.07 ^a	4.50 ± 0.11 ^b	0.89 ± 0.01 ^a	6.61 ± 0.16 ^c	1.43 ± 0.08 ^a
ΣCQA	Drip	9.03	3.3	4.23	2.52	6.51	3.02
	Mocha	7.47	1.62	7.34	2.03	8.34	2.11
	Filtered	10.23	6.73	10.35	2.64	10.16	6.7
	Infusion	8.16	1.4	6.17	1.37	9.04	2.4
di-CQA	Drip	2.55 ± 0.12 ^a	2.12 ± 0.09 ^b	0.76 ± 0.04 ^a	1.84 ± 0.11 ^b	1.76 ± 0.04 ^{a*}	1.82 ± 0.13 ^{b*}
	Mocha	2.39 ± 0.05 ^a	1.28 ± 0.07 ^a	2.18 ± 0.06 ^b	1.58 ± 0.14 ^{ab}	2.15 ± 0.03 ^b	1.40 ± 0.05 ^a
	Filtered	3.00 ± 0.08 ^b	3.70 ± 0.09 ^c	2.93 ± 0.10 ^b	1.79 ± 0.16 ^b	2.68 ± 0.09 ^c	3.18 ± 0.09 ^c
	Infusion	3.19 ± 0.16 ^b	1.40 ± 0.14 ^a	2.48 ± 0.20 ^c	1.37 ± 0.06 ^a	2.53 ± 0.06 ^c	1.98 ± 0.11 ^b
Σ(CQA + di-CQA)	Drip	11.59	5.41	4.99	4.36	8.26	4.84
	Mocha	9.86	2.9	9.52	3.6	10.49	3.52
	Filtered	13.22	10.43	13.28	4.44	12.84	9.89
	Infusion	11.35	2.81	8.65	2.74	11.58	4.38

Data denoted by the same lowercase letters are not statistically different ($\alpha = 0.05$) in terms of brewing method, and denoted by an asterisk are not different in terms of roasting degree.

The dominant chlorogenic acid in all the extracts was 3-CQA. The highest level of 3-CQA [7.51 mg (g dm)⁻¹] was found in filtered Colombian light roasted spent coffee grounds. Similar amounts of 7.30 and 7.25 mg (g dm)⁻¹ were detected in spent coffee grounds from Mexican and Nicaraguan light roasted filter coffee. On the other hand, the lowest 3-CQA concentrations were found in spent coffee grounds from Colombian and Nicaraguan dark roasted coffee brewed by infusion, at the level of 0.89 and 0.91 mg (g dm)⁻¹, respectively. It was observed, that spent coffee grounds from light roasted beans contained about 2.8 times less 5-CQA compared to 3-CQA. In the case of SCG from dark roasted beans 5-CQA was determined in the amount of about 2.7 times less than 3-CQA.

Regardless of the geographical origin of the bean and the method of brewing, the extracts from light roasted SCG were richer in 5-CQA and 3-CQA acids in comparison to the extracts from dark roasted SCG. The highest drop in 3-CQA content due to the roasting process (about 6.4 times) was observed in SCG from Nicaraguan infusion brews. The SCG from Mexican filter coffee registered the lowest drop in 3-CQA content (about 1.6 times). In the case of 5-CQA, concentration decrease was observed at levels between 1.7 and 6.3 times for SCG from Colombian drip coffee and Nicaraguan infusion brews, respectively. The decline in 3-CQA and 5-CQA acids content can be attributed to changes occurring at high temperatures during the roasting process. As a result, lactones and

other derivatives, mainly cinnamoyl-1,5- γ -quinolactones can be formed (Farah *et al.*, 2005).

All of the SCG samples were characterised by low contents of 4-CQA [below 1 mg (g dm)⁻¹]. The total content of diester derivatives of chlorogenic acids in light roasted SCG was at the level of 5-CQA acid. On the other hand, the total content of diester derivatives of chlorogenic acids in the dark roasted SCG was at the level of 3-CQA acid content. Literature data confirm the obtained results. Bravo *et al.* (2012) showed that in aqueous coffee extracts, the content of di-CQA was at a similar level as monochlorogenic acids. Similarly, Ludwig *et al.* (2012) noticed that di-CQA acids are extracted to brews much slower than monochlorogenic acids, which can result in their relatively high amounts remaining in the SCG.

Overall, SCG from filter coffee had the highest content of chlorogenic acids up to 13.28 mg 3-CQA (g dm)⁻¹ for SCG from light roasted and 10.43 mg 3-CQA (g dm)⁻¹ for SCG from dark roasted coffee. As was the case with caffeine, brewing in a filter coffee machine was the least efficient method of extraction. It is worth noting that dark roasted Nicaraguan and Mexican SCG from filter coffee machine were characterised by 1.9–3.7 times higher content of chlorogenic acids compared to other brewing methods.

The content of bioactive compounds, such as caffeine or chlorogenic acids, in coffee brews and SCG, is influenced mainly by proper selection of coffee beans and conditions of roasting and brewing. Contrary to other reports, here a uniform ratio of coffee to water, brewing time and water temperature were used to eliminate the variability associated with these factors. The use of different brewing conditions for each method may give even more varied results (Gloess *et al.*, 2013).

During the coffee roasting process, as a result of caramelisation and Maillard reactions, browning compounds, like melanoidins, are formed. These compounds are responsible for the typical aroma and the colour of roasted coffee beans and its brews (Del Pino-García *et al.*, 2012). Determination of Maillard reaction products formed during roasting process can be done by absorption measurements at a wavelength of 420 nm. At wavelengths higher than 400 nm, no other natural food compounds are known to absorb light (Cammerer *et al.*, 1999). Therefore the absorption measurement at 420 nm known as the browning index is considered to be an indirect estimation of the content of compounds resulting from browning reactions (Kim *et al.*, 2014).

Results of the browning coefficient of the coffee grounds extracts are presented in Table 3. The highest level of browning index (above 0.30) was found in SCG extracts from dark roasted Nicaraguan drip coffee. Browning index of SCG from light roasted Colombian beans prepared by infusion was the lowest of all samples and reached the level of 0.11. The

browning index of light roasted SCG extracts was 48–76% lower than the browning index value of the dark roasted SCG extracts. Regardless of the country of origin and the degree of roasting, coffee grounds obtained by infusion were characterised by the lowest level of browning index. Bravo *et al.* (2012) observed similar browning index values for *C. arabica* SCG brewed in a filter coffee machine (about 0.165). The authors also noticed that the browning index of coffee grounds extracts was 3–5 times lower than the browning index of the brews. According to Passos & Coimbra (2013), SCG comprise about 16% of browning compounds fraction. However, the chemical composition of this fraction has not been fully understood yet.

The data in Table 4 show the results of total polyphenol content and overall antioxidant activity of spent coffee grounds extracts. The total phenolics content in the SCG extracts was between 25.13 and 46.23 mg g⁻¹ of SCG dry matter in the case of light Mexican mocha and dark Nicaraguan drip spent coffee grounds respectively. In most cases, regardless of the roasting degree, the lowest content of total polyphenols was observed in the extracts from SCG obtained from Mocha and infusion brewing methods. Based on the results obtained, it can be assumed that the degree of roasting and the method of brewing can have a significant impact on the total polyphenol content in spent coffee grounds.

Phenolic compounds content in the studied SCG extracts were higher than reported by other authors (Mussatto *et al.*, 2011; Bravo *et al.*, 2012; Shang *et al.*, 2017). Values stated in the literature correspond to approximately 16.2–26.0 mg GAE g⁻¹. These differences in the concentration of total phenolics in SCG extracts may be attributed to the type and origin of coffee, roasting process parameters and extraction process conditions used. The studied SCG extracts had higher total phenolic content than the content reported in other plant materials known for their antioxidant activity such as raspberry (15.3 mg GAE g⁻¹) or blackberry (14.8 mg GAE g⁻¹) (Wang & Hsin-Shan, 2000).

It is well known that the presence of many bioactive compounds such as chlorogenic acids, caffeine and browning compounds contributes to the antioxidant activity of both coffee and spent coffee grounds (Dórea

Table 3 Browning index of SCG methanol extracts

Brewing	Nicaragua		Columbia		Mexico	
	Light	Dark	Light	Dark	Light	Dark
Drip	0.15	0.31	0.13	0.25	0.14	0.24
Mocha	0.15	0.23	0.14	0.26	0.13	0.27
Filtered	0.16	0.21	0.16	0.25	0.15	0.25
Infusion	0.12	0.19	0.11	0.21	0.12	0.19

Table 4 Total polyphenols content (TPC) [mg GA (g dm)⁻¹] and antioxidant activity (AA) of spent coffee grounds methanol extracts [mg Trolox (g dm)⁻¹]

	Brewing	Nicaragua		Columbia		Mexico	
		Light	Dark	Light	Dark	Light	Dark
TPC	Drip	39.10 ± 1.71 ^c	46.23 ± 2.70 ^c	34.48 ± 0.51 ^{b*}	34.78 ± 3.17 ^{b*}	37.87 ± 1.19 ^d	33.02 ± 1.23 ^c
	Mocha	27.78 ± 0.70 ^{a*}	26.58 ± 2.19 ^{a*}	30.38 ± 1.36 ^a	36.06 ± 2.66 ^b	25.13 ± 0.74 ^a	26.61 ± 0.96 ^a
	Filtered	35.55 ± 0.11 ^b	36.40 ± 0.85 ^b	33.10 ± 2.75 ^{ab*}	36.30 ± 0.20 ^{b*}	35.31 ± 1.06 ^{c*}	34.58 ± 1.48 ^{c*}
	Infusion	45.08 ± 1.3 ^d	28.84 ± 4.84 ^a	30.77 ± 3.31 ^{a*}	27.68 ± 1.33 ^{a*}	27.95 ± 0.59 ^b	29.64 ± 1.05 ^b
AA	Drip	65.66 ± 1.36 ^b	89.22 ± 0.84 ^d	95.24 ± 0.34 ^{d*}	97.34 ± 0.63 ^{d*}	76.36 ± 1.17 ^d	113.61 ± 0.58 ^b
	Mocha	58.90 ± 0.76 ^a	77.26 ± 0.69 ^b	67.22 ± 0.98 ^a	91.17 ± 1.26 ^c	45.06 ± 0.41 ^a	65.10 ± 1.30 ^a
	Filtered	59.78 ± 1.00 ^a	84.73 ± 1.23 ^c	76.92 ± 0.60 ^c	81.13 ± 1.01 ^b	68.75 ± 1.97 ^{c*}	65.57 ± 0.53 ^{a*}
	Infusion	93.81 ± 0.70 ^c	73.50 ± 1.22 ^a	71.80 ± 0.89 ^b	65.86 ± 0.92 ^a	53.44 ± 1.10 ^b	66.06 ± 1.08 ^a

Data denoted by the same lowercase letters are not statistically different ($\alpha = 0.05$) in terms of brewing method, and denoted by an asterisk are not different in terms of roasting degree.

& da Costa, 2005). The antioxidant activity of coffee grounds extracts ABTS ranged from 45.06 to 113.61 mg Trolox per gram of SCG dry matter. The SCG obtained by percolation brewing methods were usually characterised by both the highest antioxidant activity and highest total phenolic content. The only samples with no significant differences in total polyphenol content were Nicaraguan SCG from Mocha coffeemaker and Colombian SCG from a drip. Extracts of dark roasted SCG generally showed higher antioxidant activity than the extracts of light roasted SCG. For example SCG extracts from Mexican dark roasted beans prepared with the drip presented antioxidant activity at the level of 113 mg Trolox (g dm)⁻¹ while at the same time SCG extracts from light roasted beans (same origin and brewing method) showed lower antioxidant activity by about 37 mg Trolox (g dm)⁻¹. The increase in antioxidant activity may be a result of the higher content of browning compounds formed during roasting, which is confirmed by the results obtained during the browning index measurement. Moreover, these high values of antioxidant activity corroborated the high levels of total phenolic contents obtained in the studied extracts.

It was observed that in the present study the antioxidant activity levels were above the range reported by other authors, 10.7–53.8 mg Trolox (g dm)⁻¹ (Bravo *et al.*, 2012, 2013; Shang *et al.*, 2017). The differences between the results present in this study and the ones reported in the literature could be explained by the factors described during the total phenolic content analysis such as different agricultural and process conditions.

Conclusions

As demonstrated in this study spent coffee grounds obtained from typical household brewing methods, especially spent coffee grounds from Nicaragua origin, can be a valuable source of bioactive compounds.

Notably spent coffee grounds obtained from filter coffeemaker contain high levels of caffeine, phenolic compounds, melanoidins and possess high antioxidant activity. Extraction yield of bioactive compounds from SCG also depends heavily on the geographical origin and roasting degree of roasted coffee beans. These results indicate that SCG, which is still an underutilised coffee by-product, generated in large quantities, could potentially be considered as a promising and valuable material in different industrial areas. Both spent coffee grounds and bioactive compounds extracts can be used as a food ingredient in innovative products with high nutritional value and potential to reduce the risk of developing type 2 diabetes and obesity. However, in this study solid-liquid extraction of bioactive compounds was performed using methanol as an extraction solvent. Due to its high polarity and high extraction yield it is recognised as the most efficient and economical solvent for extraction of bioactive compounds from plant materials. The main detrimental characteristic of methanol, which raises serious issues with the potential application of the extracts in the industries demanding high levels of safety such as food or pharmaceutical industry, is its high toxicity. Because of the aforementioned reasons, it is clear that the next step of the research should be finding an alternative and safe method of extracting bioactive compounds from spent coffee grounds with extraction yields comparable to solid-liquid extraction with methanol as a solvent.

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